Lily L. Zhao NASA Sagan Fellow | University of Chicago

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Education

Yale University
M.S., M.Phil., Astronomy — May 2018
Ph.D., Astronomy — Jun. 2021
Dissertation Title: *The Path to Extreme Precision Radial Velocity With EXPRES*

University of Chicago — Jun. 2016
B.S. Mathematics
B.A. Physics
B.A. Biological Sciences

Research Positions

University of Chicago — Chicago, IL
NASA Sagan Fellow — Oct. 2024 - Present

Center for Computational Astrophysics, Flatiron Institute — New York, NY
Flatiron Research Fellow — Sep. 2021 - Sep. 2024
Pre-Doctoral Fellow — Sep. 2019 - Jan. 2020

Yale Exoplanet Group — New Haven, CT
NSF Graduate Research Fellow — Sep. 2016 - Jun. 2021

Awards

Dirk Brouwer Memorial Prize for Outstanding PhD Thesis, Yale University — 2024
Hubble Fellowship, NASA — 2024
Third Place, Three Minute Thesis Competition, Yale University — 2020
Sheldon Wise Pre-Doctoral Fellowship, Yale University — 2018
Graduate Research Fellowship, National Science Foundation — 2016

Publications

First Author

10. **Zhao, L.L.**, Al Moulla, K., Faria, J., et al. "The Extreme Stellar–Signals Project IV. State of the Field of Disentangling Solar Signals" in prep.
9. **Zhao, L.L.**, Fischer, D.A., Szymkowiak, A.E., et al. "Uncovering Hidden Systematics in Extreme-Precision Radial Velocity Measurements" 2026, ApJS, 282, 71
8. **Zhao, L.L.**, Bedell, M., Hogg, D.W., Luger, R. "A Compact, Coherent Representation of Stellar Surface Variation in the Spectral Domain" 2024, ApJ, 977, 140
7. **Zhao, L.L.**, Dumusque, X., Ford, E., et al. "The Extreme Stellar–Signals Project III. Combining Solar Data from HARPS, HARPS–N, EXPRES, and NEID" 2023, AJ, 166, 173
6. **Zhao, L.L.**, Kunovac-Hodzic, V., Brewer, J.M., et al. "Measured Spin-Orbit Alignment of Ultra-Short Period Super-Earth 55 Cnc e" 2023, Nature Astronomy, 7, 198
5. **Zhao, L.L.**, Fischer, D.A., Henry, G.W., et al. "The EXPRES Stellar-Signals Project II. State of the Field of Disentangling Photospheric Velocities" 2022, AJ, 163, 171
4. **Zhao, L.L.**, Hogg, D.W., Bedell, M., Fischer, D.A. "*Excalibur*: A Non-Parametric, Hierarchical Wavelength-Calibration Method for a Precision Spectrograph" 2021, AJ, 161, 80

3. **Zhao, L.L.**, Fischer, D.A., Ford, E., Henry, G.W., Rottenbacher, R.M., Brewer, J.M. "The EXPRES Stellar-Signals Project I. Description of Data" 2020, RNAAS, 4, 156
2. Petersburg, R.R., Ong, J.M.J., **Zhao, L.L.**, et al. "An Extreme-Precision Radial-Velocity Pipeline: First Radial Velocities from EXPRES" 2020, AJ, 159, 187 (Contributions were equally split among the first three authors)
1. **Zhao, LL..**, Fischer, D., Brewer, J., Giguere, M., & Rojas-Ayala, B. "Planet Detectability in the Alpha Centauri System" 2018, AJ, 155, 24

Contributing Author ($\star$: Student Paper)

25. Luhn, J., Rubenzahl, R.A., Halverson, S., **Zhao, L.L.** "An Exposure-averaged Gaussian Process Framework to Recover Stellar Variability in Combined Radial Velocity Data Sets" arXiv, 2601.02462
24. Sikora, J., Llama, J., Roettenbacher, R., et al. [incl. textbfZhao, L.L.] "Confirmation of Fe I on MASCARA-5 b's Dayside Observed With EXPRES" 2026, AJ, 171, 322
23. Brady, M., Bean, J.L., Basant, R., et al. [incl. **Zhao, L.L.**] "An Earth-like Density for the Temperate Earth-sized Planet GJ 12b" 2025, AJ, 170, 335
22. Ellworth, M., Llama, J., **Zhao, L.L.**, et al. "The He I D3 Line as a Proxy for Magnetic Activity using EXPRES Solar Observations" 2025, AJ, 994, 104
$\star$21. Komori, C., Brewer, J.M., **Zhao, L.L.** "The Effects of Sunspots on Spectral Line Shapes in the Visible" 2025, AJ, 170, 209
20. Salzer, J., Cisewski-Kehe, J., Ford, E.B., **Zhao, L.L.** "Searching for Low-Mass Exoplanets Amid Stellar Variability with a Fixed Effects Linear Model of Line-by-Line Shape Changes" 2025, AJ, 170, 179
19. Freckelton, A.V., Mortier, A., Bedell, M., et al. [incl. **Zhao, L.L.**] "gr8stars – I. A homogeneous spectroscopic study of bright FGKM dwarfs and a public library of their high-resolution spectra" 2025, MNRAS, 540, 1786
18. Basant, r., Luque, R., Bean, J.L., et al. [incl. **Zhao, L.L.**] "Four sub-Earth planets orbiting Barnard's Star from MAROON-X and ESPRESSO" 2025, ApJ, 982, 1
17. Vieytes, M., **Zhao, L.L.**, Bedell, M."The Influence of Chromospheric Activity on Line Formation" 2025, ApJ, 981, 4
16. Savel, A.B., Bedell, M., Kempton, E.M-R., et al. [incl. **Zhao, L.L.**] "Peering into the black box: forward-modeling the uncertainty budget of high-resolution spectroscopy of exoplanet atmospheres" 2025, AJ, 169, 135
$\star$15. Lam, C., Bedell, M., **Zhao, L.L.**, Gupta, A. "*Gaspery*: Optimized Scheduling of Radial Velocity Follow-Up Observations for Active Host Stars" 2024, AJ, 168, 200
14. Llama, J., **Zhao, L.L.**, Brewer, J.M., et al. "The Lowell Observatory Solar Telescope: a fiber feed into the Extreme Precision Spectrometer " 2024, SPIE, 13094, 20L
13. Siegel, J., Halverson, S., Luhn, J.K., **Zhao, L.L.**, et al. "Quiet Please: Detrending Radial Velocity Variations from Stellar Activity with a Physically Motivated Spot Model" 2024, AJ, 168, 158
12. Eisner, N.L., Grunblat, S.K., Barragán, O., et al. [incl. **Zhao, L.L.**] "Planet Hunters TESS. V. A Planetary System Around a Binary Star, Including a Mini-Neptune in the Habitable Zone" 2024, AJ, 167, 241
11. Korolik, M., Rottenbacher, R.M., Fischer, D.A., et al. [incl. **Zhao, L.L.**] "Refining the Stellar Parameters of τ Ceti: a Pole-on Solar Analog " 2023, AJ, 166, 123
10. Brewer, J.M., **Zhao, L.L.**, Fischer, D.A., et al. "EXPRES IV. Two Additional Planets Orbiting ρ Corona Borealis Reveal Uncommon System Architecture" 2023, AJ, 166, 46
9. Rottenbacher, R.M., Cabot, S.H.C., Fischer, D.A., et al. [incl. **Zhao, L.L.**] "EXPRES. III. Revealing the Stellar Activity Radial Velocity Signature of ϵ Eridani with Photometry and Interferometry" 2021, AJ, 163, 19
8. Luger, R., Bedell, M., Foreman-Mackey, D., et al. [incl. **Zhao, L.L.**] "Mapping Stellar Surfaces III: An Efficient, Scalable, and Open-Source Doppler Imaging Model" 2021, arXiv:2110.06271

7. Holzer, P., Cisewski-Keke, J., Fischer, D.A., **Zhao, L.L.** "A Hermite-Gaussian Based Radial Velocity Estimation Method" 2021, AnApS, 15, 527
6. Holzer, P.H., Cisewski-Kehe, J., **Zhao, L.L.**, Fischer, D.A., Ford, E.B. "A Stellar Activity F-statistic for Exoplanet Surveys (SAFE)" 2021, AJ, 161, 272
5. Cabot, S.H.C., Roettenbacher, R.M., Henry, G.W., **Zhao, L.L.**, et al. "*EXPRES*. II. Searching for Planets Around Active Stars: A Case Study of HD 101501" 2020, AJ, 161, 26
4. Hoeijmakers, H.J., Cabot, S.H.C., **Zhao, L.L.**, et al. "High-Resolution Transmission Spectroscopy of MASCARA-2 b with EXPRES" 2020, A&A, 641, A120
3. Brewer, J.M., Fischer, D.A., Blackman, R.T., et al. [incl. **Zhao, L.L.**] "EXPRES I. HD 3651 an Ideal RV Benchmark" 2020, AJ, 160, 67
2. Blackman, R.T., Fischer, D.A., Jurgenson, C.A., et al. [incl. **Zhao, L.L.**] "Performance Verification of the EXtreme PREcision Spectrograph" 2020, AJ, 159, 238
1. Gaudi, S., Blackwood, G., Howard, A., et al. [incl. **Zhao, L.L.**] "Extreme Precision Radial Velocity Working Group" 2019, BAAS 51, 232

Textbooks

Astrobiology (Pressbooks) 2024
Co-author

Handbook of Exoplanets (Springer) 2023
55 Cancri (Copernicus): A Multi-planet System with a Hot Super-Earth and a Jupiter Analogue

Origins and the Search for Life in the Universe (CK-12) 2017
Chapter 6: The Complexification of Chemistry
Chapter 7: The Emergence of Life on Earth

INVITED TALKS

Colloquia

- CU Boulder (Apr. 2026)
- Columbia University (Feb. 2026)
- University of Toledo (Feb. 2026)
- University of Texas at Austin (Feb. 2026)
- UC Berkeley (Feb. 2025)
- Rochester Institute of Technology (Dec. 2024)
- University of Toronto (Feb. 2024)
- University of Maryland (Apr. 2023)
- Jet Propulsion Laboratory (Nov. 2022)
- EPRV Research Coordination Network (May 2022)

Seminars Not listed: 12 seminars (9 invited)

Invited Conference Talks

- "Advancing Precision Radial Velocity Towards Detecting Earth Analogs" *Frank N. Bash Symposium* (Sep. 2025)
- "Advancing Precision Radial Velocity Towards Detecting Earth Analogs" *TDLI: Postdoctoral Frontier Symposium in Physics and Astronomy* (Sep. 2025)
- Life Beyond Earth: The Missing Links *ISSI Breakthrough Workshop* (Jun. 2025)
- "Solar to Stellar Observations" *Cool Stars* (Jun. 2024)
- "Excalibur" *Spectral Fidelity* (Sep. 2023)
- "Comparing Solar Data across Four Precision Instruments" *PoET* (Feb. 2023)
- "Improving Exoplanet Detection with Discriminative Linear Regression" *Flatiron-wide Algorithms and Mathematics* (Oct. 2022)

- "Machine Learning for Extreme Precision Spectrographs" *AAS 238; Machine Learning in Astronomy (MiM)* (Jun. 2021)
- "EXPRES" *Extreme Precision Radial Velocity IV* (Mar. 2019)

Not listed: 21 contributed conference talks

PROFESSIONAL ACTIVITIES

Referee: AAS Journals, A&A, MNRAS, PASP, PASJ
Proposal Reviewer: NASA APRA, NSF AAG, NSF ATI

Community Leadership & Collaborations

Exoplanet Exploration Program Analysis Group (ExoPAG) 2023 - 2026
Executive Committee Member

EPRV Research Coordination Network 2022 - Present
Steering Committee Member

The Terra-Hunting Experiment 2021 - Present
Member

Extreme Stellar Signals Project (ESSP) 2020 - Present
Founder; Executive Committee Member

Extreme Precision Spectrograph (EXPRES) Team 2016 - Present
Project Scientist

Scientific Organizing Committees

EPRV VI Splinter Session "Accessing and Working with EPRV Solar Datasets" 2025
EPRV VI Splinter Session "Benchmarking Stellar Variability Mitigation Methods" 2025
EPRV V 2023
EPRV V Splinter Session "From Solar to Stellar" 2023
Sun-as-a-Star Workshop 2023
Emerging Researchers in Exoplanet Science (ERES)
- ERES III, Yale 2017
- ERES V, Cornell 2019
- ERES VI, Princeton 2021

Diversity, Inclusion, & Equity

Executive Board: Yale Astronomy Climate and Diversity Committee 2020 - 2021
Fellow: Yale Office of Graduate Student Diversity and Development 2018 - 2021
Founding Member: Yale Astronomy Student Council 2018 - 2021

Invited Panels

Interview Preperation Panel, Guide to Applying for Astronomy Postdocs (GAAP) (Jul. 2025)
Career Panel, Emerging Researchers in Exoplanet Science IX (Jul. 2024)

MENTORING

Co-Mentor: Claire Komori 2023-2025
Masters Student, State University of San Francisco
First-author publication

Co-Mentor: Chris Lam Fall 2022
Graduate Student, University of Florida
First-author publication, Poster presentation at EPRV V

Mentor: Nusrat Jahan Summer 2022
Undergraduate Student, Hunter College
Poster presentation at AAS 241 and CUWiP

Mentor: Lianys Feliciano — Summer 2022
Undergraduate Student, New York City College of Technology
Poster presentation at SACNAS and AAS 241

TEACHING

Guest Lecture: Another Earth — Fall 2022
Columbia University

Research Project Lead: Exoplanets — Summer 2021
Warrior Scholars Project

Certificate of College Teaching Preparation — Awarded 2018
Granted by the Yale Center for Teaching and Learning

Co-Instructor: Origins and the Search for Life in the Universe — Fall 2017
Yale University

Teaching Fellow: Frontiers and Controversies in Astrophysics — Spring 2017
Yale University

Teaching Fellow: Origins and the Search for Life in the Universe — Fall 2016
Yale University

SELECT OUTREACH

Astronomy Conversations Presenter: Adler Planetarium	2025 - Present
Mentor: AMP-UP	2024 - 2025
Reviewer: NHFP App Feedback Program	2024
Speaker: Skype a Scientist	2019 - 2021
Docent: the Peabody Museum	2018 - 2019
Demonstrations, Group Leader: Girls Science Investigation	2017 - 2019
Guest Author: Scientific American, Observations	2017
Invited Speaker, Public Relations Committee: Open Labs	2016 - 2020
Observatory Volunteer: Franklin Institute	2012 - 2016

PROPOSALS

Observing Proposals

PI: NEID, 2022B
Awarded 5.8 hours of P2 time
"Measuring the Shortest Timescale Stellar Signals for a Range of Spectral Types"

PI: Gemini, 2022B
Awarded 29.8 hours of Band 1 time
"Unveiling the Signatures of Starspots in MAROON-X Spectra with Simultaneous Interferometric Stellar Surface Mapping"

Grant Proposals

While fully funded from 2021-2027, I contributed to the following successful proposals.

Co-I: NASA Exoplanet Mass Measurement Program — 2024
Selectable
"A Star-as-the-Sun: applying solar EPRV techniques to stellar spectra"
(PI: Drake Demming, University of Maryland)

Co-I: NASA Exoplanet Mass Measurement Program — 2024
Selectable
"Development of a community data reduction framework to advance EPRV towards the cm/s precision era"
(PI: Daniel Krolikowski, University of Arizona)

Co-I: NASA Extreme Precision Radial Velocity Foundation Science 2022
Awarded $450,000
"New Strategies for Combining EPRV Observations from Multiple Instruments" (PI: Eric Ford, Pennsylvania State University)

Co-I: NASA Extreme Precision Radial Velocity Foundation Science 2022
Awarded $480,000
"A community driven, modular data-pipeline architecture to push EPRV into the 1 cm/s era" (PI: Jennifer Burt, California Institute of Technology)

Collaborator: NSF Astronomical Sciences 2023 - 2025
Awarded $510,000
"Unmasking Stellar Variability: Hierarchical Bayesian methods for characterization of low-mass planets with EPRV spectroscopy" (PI: Jessica Kehe, University of Wisconsin-Madison)

Co-I: NASA Exoplanets Research Program (XRP) 2023 - 2025
Awarded $575,000
"Turn down the noise! Disentangling planetary and stellar signals by observing the Sun with EXPRES" (PI: Joe Llama, Lowell Observatory)

Co-I: Heising-Simons Foundation 2022 - 2025
Awarded $950,000
"EXPRES 100 Earths Survey" (PI: Joe Llama, Lowell Observatory)

References

Debra A. Fischer: debra.fischer@yale.edu
David W. Hogg: david.hogg@nyu.edu
Eric B. Ford: eford@psu.edu